

WR-VII

Energy–Time Frontiers of World Realizability

Calibrated quantum responses, action bounds
and generator ambiguity
Mathematical draft v0.2

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Abstract

A boundary in an energy–time diagram requires a declared physical evolution, calibrated responses and a resource with specified units. WR-VII supplies such an interface for finite-dimensional closed unitary dynamics, with energy resource defined as Hamiltonian spectral diameter. A self-contained projector-response argument gives an action certificate and a sharp lower frontier for an observed transition probability. A constant-qubit benchmark then computes the entire compatible generator class, including separated revival windows. At the resource frontier, transition probability selects the spectral diameter but leaves a circle of distinct Hamiltonians. Additional phase-calibrated responses identify this circle for a nonorthogonal final state; at complete transfer, even final-state tomography cannot identify the generator, while an intermediate-time protocol can. Response intervals and independent finite repetitions give separate deterministic and confidence-qualified feasibility certificates. The arguments use established quantum speed-limit, Hamiltonian-identification and concentration tools, with no theorem-priority claim. They certify a boundary of the declared response-and-control model, not an upper energy of reality, a universal lifetime or a cosmological realizability wall.

Keywords: energy–time frontier; spectral diameter; quantum response; generator completion; finite confidence; protocol refinement. Parallel EN/RU texts share all result and equation numbers.

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1 The new obligation: operational frontiers

WR-I defines response equivalence and identifiable quantities [1]; WR-II separates finite consistency from full global realization [2]; WR-III studies nonempty response fibres [3]; WR-IV distinguishes complete realizations from history projections [4]. WR-V adds experimental transition descriptions [5], and WR-VI records measured data, structural assumptions and separately justified hard bounds [6]. The WR-VI activity cap is dimensionless. Its numerical threshold cannot acquire units of energy or duration by renaming a parameter.

The new obligation is to construct a duration-bearing quantum response model and a resource-feasibility region, then ask which generator parameters that region identifies. Classical transition kernels and coherent unitary generators remain distinct carriers. The inherited interface is compatibility with calibrated responses and declared restrictions. Quantum speed limits and Hamiltonian identification already have extensive prior art [7, 8]; the contribution here is an explicit, auditable WREH completion calculation, with elementary proofs and stated limitations.

A lower resource frontier asks when one specified response can be produced within a calibrated model. A global realizability boundary would require an independently declared class of complete worlds and a justified response map. These questions have different domains. No upper energy of the Universe or maximum lifetime follows from the construction below.

2 Carrier, clock and energy record

Let $\mathcal{H} = \mathbb{C}^n$, $n \geq 2$. On the calibrated interval $[0, \tau]$, $\tau > 0$, a Hermitian Hamiltonian $H(t)$ with integrable operator norm generates the normalized absolutely continuous solution

$$i\hbar\dot{\psi}(t) = H(t)\psi(t), \quad \|\psi(0)\| = 1. \quad (1)$$

For a fixed orthogonal projector Q , prepare $Q\psi(0) = 0$ and measure its final Born probability. Define

$$p(t) = \langle \psi(t), Q\psi(t) \rangle, \quad D(t) = \lambda_{\max}(H(t)) - \lambda_{\min}(H(t)), \quad A = \frac{1}{2\hbar} \int_0^\tau D(t) dt. \quad (2)$$

D and its pointwise upper cap $E \geq 0$ have units of joules; τ has units of seconds; A is dimensionless. Spectral diameter is invariant under $H(t) \mapsto H(t) + c(t)I$. Such a shift changes only a global state phase. The resource is neither consumed work nor mean energy, and is not the state's energy standard deviation. A claim involving those quantities needs a different record. The finite-dimensional closed-system law, clock, preparation, measurement and hard diameter cap are supplied assumptions that must be calibrated in an apparatus.

Definition 2.1 (Operational completion class). *With the preceding carrier and protocol fixed, $\mathcal{C}(E, \tau, J)$ consists of Hamiltonian histories satisfying $D(t) \leq E$ almost everywhere and $p(\tau) \in J$, where $J \subseteq [0, 1]$ is the supplied response set. A constant-qubit subclass will be stated separately.*

Nonemptiness says that at least one allowed generator produces the response. It does not require every allowed generator to do so, or identify the actual generator. The evolution duration excludes preparation, readout and resetting overhead. An inference about total experimental time must additionally account for repetitions and those durations.

3 A sharp response-action certificate

Theorem 3.1 (Projector-response action bound). *Under (1)–(2),*

$$\arcsin \sqrt{p(\tau)} \leq A. \quad (3)$$

Proof. Set $u = Q\psi$ and $v = (I - Q)\psi$. Their norms are $\sqrt{p}$ and $\sqrt{1-p}$, and $u \perp v$. For $c = (\lambda_{\max}H + \lambda_{\min}H)/2$, $\|H - cI\| = D/2$. The commutator formula therefore gives, almost everywhere,

$$|\dot{p}| \leq \frac{2}{\hbar} |\langle u, (H - cI)v \rangle| \leq \frac{D}{\hbar} \sqrt{p(1-p)}. \quad (4)$$

$$F_\eta(x) = \int_0^x \frac{ds}{2\sqrt{(s+\eta)(1-s+\eta)}}, \quad \eta > 0.$$

To handle $p = 0, 1$ without dividing by zero, for $\eta > 0$ use the regularized antiderivative displayed above. Its derivative is bounded on $[0, 1]$ and the absolutely continuous chain rule gives $|dF_\eta(p)/dt| \leq D/(2\hbar)$. Integrate from $p(0) = 0$, then let $\eta \downarrow 0$. Monotone convergence of this nonnegative integrand gives $F_\eta(x) \rightarrow \arcsin \sqrt{x}$, including both endpoints. $\square$

Corollary 3.2 (Sharp energy–duration frontier). *If $D(t) \leq E$, an exact response $p \in [0, 1]$ requires $E\tau \geq 2\hbar \arcsin \sqrt{p}$. Conversely, on a qubit with $\psi(0) = |0\rangle$ and $Q = |1\rangle\langle 1|$, this condition is sufficient. Its full reachable probability set is*

$$\mathcal{R}(E, \tau) = [0, \sin^2(\min\{C, \pi/2\})], \quad C = \frac{E\tau}{2\hbar}. \quad (5)$$

Proof. Necessity follows from $A \leq C$. For sufficiency set $\theta = \arcsin \sqrt{p}$ and $H = (\hbar\theta/\tau)\sigma_x$. Its diameter is $2\hbar\theta/\tau \leq E$ and its final transition probability is $\sin^2 \theta = p$. For $p = 0$ use $H = 0$. Every probability in (5) has this construction; none above it satisfies the action bound. $\square$

This is a quantum speed certificate in a supplied unitary model, related to the established geometric energy–time setting [7]. At complete transfer it gives $E\tau \geq \pi\hbar$. It is not a general energy–time product law for arbitrary physical objects. Sharpness uses a freely chosen transverse qubit generator; a hardware control alphabet or inaccessible axes can make the actual attainable set smaller.

For a time-independent H and orthogonalization, the Mandelstam–Tamm and Margolus–Levitin bounds use, respectively, the state’s energy spread ΔH and mean energy above ground $\bar{\varepsilon} = \langle H \rangle - \lambda_{\min}(H)$ [11, 12]. For positive denominators they read

$$\tau_\perp \geq \frac{\pi\hbar}{2\Delta H}, \quad \tau_\perp \geq \frac{\pi\hbar}{2\bar{\varepsilon}}.$$

Since $\Delta H \leq D/2$, a state-specific spread can give a stronger bound than the diameter-only orthogonalization certificate. The mean-energy bound is stronger than $\pi\hbar/D$ when $\bar{\varepsilon} < D/2$; it does not uniformly improve it. For the transverse frontier family at complete transfer, $\Delta H = \bar{\varepsilon} = E/2$ and the three bounds agree. These state resources are not supplied inputs to Theorem 3.1, and the static mean-energy formula is not applied to arbitrary $H(t)$. Corollary 3.2 is sharp over its freely controlled diameter-constrained class, not an optimal time estimate for every fixed state and Hamiltonian.

4 Complete constant-qubit generator fibre

Fix the computational basis and calibrated laboratory axes. Remove the scalar identity term, and write every nonzero constant traceless Hermitian qubit generator as

$$H = \frac{d}{2} \mathbf{n} \cdot \boldsymbol{\sigma}, \quad 0 < d \leq E, \quad \|\mathbf{n}\| = 1, \quad \phi = \frac{d\tau}{2\hbar}, \quad (6)$$

$$U(\tau) = \cos \phi I - i \sin \phi \mathbf{n} \cdot \boldsymbol{\sigma}.$$

Here d is spectral diameter, not Hilbert-space dimension. In this section the Hilbert space is fixed as $\mathbb{C}^2$. For preparation $|0\rangle$ and the $|1\rangle$ projector, direct multiplication gives

$$p = (1 - n_z^2) \sin^2 \phi. \quad (7)$$

Theorem 4.1 (All compatible constant generators). *For $0 < p < 1$, let $\theta = \arcsin \sqrt{p}$. The constant-generator completion class is parameterized by*

$$\begin{aligned} \phi &\in \left(\bigcup_{m \geq 0} [m\pi + \theta, (m+1)\pi - \theta] \right) \cap [0, C], \\ n_z &= \pm \sqrt{1 - \frac{p}{\sin^2 \phi}}, \quad (n_x, n_y) = \frac{\sqrt{p}}{|\sin \phi|} (\cos \beta, \sin \beta). \end{aligned} \quad (8)$$

with $\beta \in \mathbb{R}/(2\pi\mathbb{Z})$ and $d = 2\hbar\phi/\tau$. At window endpoints the two signs coincide. The number of nonempty separated phase windows is zero for $C < \theta$, and otherwise $1 + \lfloor (C - \theta)/\pi \rfloor$.

Proof. Equation (7) requires $\sin^2 \phi \geq p$ and fixes n_z^2 . On each nonnegative half-period, this inequality is exactly the displayed interval. The transverse magnitude follows from unit normalization, and its azimuth is unrestricted. Conversely these parameters satisfy (7) and $d \leq E$. A window is nonempty exactly when $m\pi + \theta \leq C$. Distinct windows are separated by forbidden phase gaps, so the stated count follows. $\square$

Corollary 4.2 (A frontier can retain generator ambiguity). *For $0 < p \leq 1$ and $C = \theta$, the constant-qubit class is the circle*

$$H_\beta = \frac{E}{2} (\cos \beta \sigma_x + \sin \beta \sigma_y), \quad \beta \in \mathbb{R}/(2\pi\mathbb{Z}). \quad (9)$$

Thus the exact frontier identifies $d = E$ in this subclass, but not H .

Proof. At $C = \theta$ the only allowed phase is θ , and (7) forces $n_z = 0$. Each distinct azimuth gives a distinct traceless operator. For $p = 1$, $\theta = \pi/2$ and the same argument applies directly. $\square$

This is a pointwise instance of the WR-I distinction between an identifiable quantity and its full carrier object [1]: $H \mapsto D(H)$ is constant on this constrained exact-response fibre, while H itself is not. It does not assert that diameter is identifiable on every response fibre of the entire generator carrier. Here the minimum resource is attained, by a circle of generators. This exact-data degeneracy differs from WR-II v0.5, Example 10.19 [2], and WR-VI v0.2, Corollary 5.3 [6], where a nonclosed admissible class has an unattained infimum. In WR-VI the additional irreducibility requirement excludes the minimizing matrix. No analogous strict exclusion is imposed on the present constant-qubit frontier family.

The azimuth is not declared a physical gauge: the laboratory x, y axes are fixed and additional calibrated probes can distinguish it. Only scalar identity shifts were removed. For $p = 1$, allowed phases are the points $\pi/2 + m\pi$, each with a transverse circle. For $p = 0$, phases $m\pi$ allow every axis, while other phases require $n_z = \pm 1$. At $d = 0$, all axis labels represent the same zero operator. These endpoint cases must not be obtained by dividing (8) by zero.

5 Which additional protocol identifies the circle?

For the frontier family (9), calibrated Pauli means in fresh preparations at the same duration are

$$(r_x, r_y, r_z) = (\sin \beta \sin 2\theta, -\cos \beta \sin 2\theta, \cos 2\theta). \quad (10)$$

Theorem 5.1 (State identification differs from generator identification). *Restrict to the constant frontier family. If $0 < p < 1$, exact r_x, r_y at τ identify β modulo 2π , hence H . If $p = 1$, the complete final density operator is the same for every β , so no final-state observable at that duration identifies H . At $p = 1$, exact r_x, r_y on separate preparations evolved for $\tau/2$ identify β .*

Proof. Direct multiplication of $U|0\rangle = (\cos \theta, -ie^{i\beta} \sin \theta)^T$ gives (10). For $0 < p < 1$, $s = \sin 2\theta = 2\sqrt{p(1-p)} > 0$, so $(r_x/s, -r_y/s) = (\sin \beta, \cos \beta)$ determines the azimuth. For $p = 1$, $\theta = \pi/2$, and the density operator is $|1\rangle\langle 1|$ independently of the azimuth. At half-duration the same constant generator has phase $\pi/4$, giving $(r_x, r_y) = (\sin \beta, -\cos \beta)$. $\square$

The half-duration measurement is performed on another preparation, rather than inserted into the original evolution. Exact additional means are new data, not values inferred from the first transition probability. The constant-generator and exact-frontier assumptions are essential; this is not a complete identification theorem for arbitrary time-dependent controls. Division by $2\sqrt{p(1-p)}$ also reveals a finite-resolution difficulty near both response endpoints. A finite tomography design requires its own sampling and calibration analysis.

6 Calibrated numerical benchmark and response intervals

Write $\nu = E/h$, where $h = 2\pi\hbar$. With a supplied spectral-gap cap $\nu = 10$ MHz and target $p = 1/4$, $\theta = \pi/6$ and

$$\tau_{\min} = \frac{\theta}{\pi\nu} = \frac{1}{6\nu} = \frac{50}{3} \text{ ns}, \quad E = h\nu = 6.62607015 \times 10^{-27} \text{ J}. \quad (11)$$

The exact SI value $h = 6.62607015 \times 10^{-34} \text{ J s}$ is supplied by the defining-constant convention [10]. The numerical frequency is a chosen calibrated-model input, not a measurement reported from a device. At the frontier, H_0 and $H_{\pi/2}$ have the same transition probability but different (r_x, r_y) : $(0, -\sqrt{3}/2)$ and $(\sqrt{3}/2, 0)$. Complete transfer at the same cap takes at least 50 ns, retains the same circle, and is resolved by the proposed probes at 25 ns.

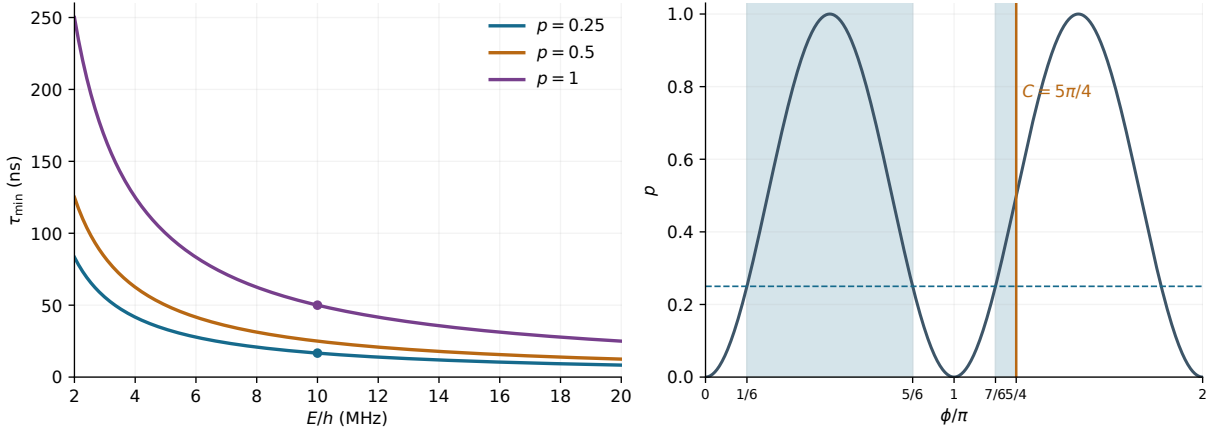


Figure 1: Left: conditional resource frontiers for three responses, with spectral diameter expressed as E/h and duration in nanoseconds. Right: allowed phases for $p = 1/4$; a larger resource budget can admit a second separated window. These are analytic model curves, not experimental observations.

Proposition 6.1 (Closed response intervals have a lower-endpoint frontier). *On the qubit carrier of Corollary 3.2, let $J = [p_-, p_+] \subseteq [0, 1]$ be nonempty. Then*

$$\mathcal{C}(E, \tau, J) \neq \emptyset \iff E\tau \geq 2\hbar \arcsin \sqrt{p_-}. \quad (12)$$

Proof. Any compatible $p \in J$ is at least p_- , so necessity follows from the monotone action certificate. For sufficiency, the constant transverse construction at p_- belongs to the class. In particular, $p_- = 0$ gives the zero generator for every budget. $\square$

The quantifier is existence of one probability in the observed interval, not one generator that simultaneously produces all probabilities. A deterministic interval has no statistical coverage by definition. A nonzero lower endpoint gives a robust lower certificate within this observation record; if zero is allowed, that certificate disappears.

7 Finite repetitions and qualified exclusion

For a prespecified duration and generator stable across fresh preparations, suppose recorded binary outcomes $X_1, \dots, X_N$ are independent and identically distributed Bernoulli variables with mean μ . Calibration supplies a uniform bound $|\mu - p(\tau)| \leq \delta$, $\delta \geq 0$, including any stated preparation/readout bias. Fix N and $0 < \alpha < 1$ before inspecting outcomes. Define

$$\hat{p} = \frac{1}{N} \sum_i X_i, \quad r_N = \sqrt{\frac{\log(2/\alpha)}{2N}}, \quad J_N = [\max\{0, \hat{p} - r_N - \delta\}, \min\{1, \hat{p} + r_N + \delta\}]. \quad (13)$$

Proposition 7.1 (Confidence-qualified resource certificate). *With probability at least $1 - \alpha$, $p(\tau) \in J_N$. On that event every compatible closed unitary model obeys $E\tau \geq 2\hbar \arcsin \sqrt{\max\{0, \hat{p} - r_N - \delta\}}$. Declaring the stated model infeasible only when its cap violates this inequality has false-exclusion probability at most α , under the supplied sampling and calibration assumptions.*

Proof. For one Bernoulli variable set $g(s) = \log \mathbb{E} e^{s(X-\mu)}$. Then $g(0) = g'(0) = 0$ and $g''(s)$ is the variance of a tilted Bernoulli law, at most $1/4$, hence $g(s) \leq s^2/8$. Independence and the exponential Markov inequality give each tail at deviation r a bound e^{-2Nr^2} after optimizing $s = 4r$ (and using the negative sign for the lower tail). The two-tail bound is $2e^{-2Nr^2}$, the Bernoulli case of Hoeffding's inequality [9]. At $r = r_N$ it is α . The bias bound enlarges the interval by δ , and Theorem 3.1 gives the resource conclusion. Under a true capped model, exclusion can occur only outside the coverage event. $\square$

This is a conservative coverage construction, not a necessary sample-size law or an optimal discrimination protocol. Adaptive stopping and drift require additional analysis. Increasing N reduces the sampling term but not the supplied bias floor. For $N = 512$, 128 recorded successes, $\alpha = 0.05$ and $\delta = 0.01$, the lower endpoint is approximately 0.17998; with $E/h = 10$ MHz its time certificate is approximately 13.95 ns. These are synthetic benchmark counts. At least $N\tau$ of evolution time is spent in sequential repetitions, with overhead added separately; parallel hardware changes that accounting.

8 Physical boundaries and global realizability

The attainable-response boundary and an internal structural wall are distinct WR-III objects [3]. For fixed E, τ with $0 < C < \pi/2$, declare the response manifold $M = (0, 1)$. Equation (5) gives the full-dimensional image $\mathcal{R}(E, \tau) \cap M = (0, \sin^2 C]$, whose boundary in M is its upper endpoint. For a target $p \in M$, this upper-boundary condition rewrites as the energy–duration frontier in Corollary 3.2. This is a response-boundary identification with a specified ambient space. It is not an assertion of failure of local fibre triviality; such a structural-wall theorem would need a separately declared total carrier, topology and response map.

The founding WR-VII task distinguishes several kinds of scales. They cannot be substituted for one another by plotting them on the same axes. The present construction discharges one explicit quantum response obligation.

Quantity	Required model	Status here
Response speed	Closed unitary evolution, diameter cap, prepared state and projector.	Conditional sharp certificate and qubit completion.
Object or vacuum lifetime	Decay law, survival response, admissibility and environment.	Separate obligation; not evolution duration τ .
Effective-theory cutoff	A declared theory and evidence for its domain of validity.	Not a global-world existence certificate.
Planck or gravitational scale	Gravitational dynamics, units, geometry and response map.	No gravitational or collapse model constructed.
Global realizability	Complete-world carrier, consistency conditions and justified lift.	Inherited WR-II gate; not supplied by a speed limit.

The manifesto's candidate suprema such as $\sup\{E : \mathcal{W}_E \neq \emptyset\}$ are meaningful only after the world class $\mathcal{W}_E$ is independently specified. Our E is an upper control cap: increasing it enlarges the feasible generator class. For a fixed reachable target and duration, feasible caps form an upper ray, so their supremum is unbounded. This is neither a calculation of the manifesto's E_* nor a refutation of a differently constructed physical cutoff. A failure of the certificate can reject the combination of calibration, preparation, dynamics and resource bound. It need not reject physical existence. An open system, missing control terms, imperfect clock or underestimated bias can violate the adopted record. Generator selection remains a projected inference: even a uniquely identified H can have multiple declared full-world lifts. The WR-I/IV projection distinction remains in force.

9 Provenance, reproduction and downstream work

The action inequality is a self-contained finite-dimensional speed-limit argument, adjacent to the established geometric framework [7]; no discovery of quantum speed limits is claimed. The qubit windows and frontier circle follow from the elementary Pauli exponential. The additional-probe calculation is a specific identification example, within established Hamiltonian-learning questions [8], not their general solution. The interval certificate is monotonicity with a sharp qubit witness. The finite-repetition result is the elementary Bernoulli concentration bound [9]. No exact WREH theorem is attributed to these sources, and exhaustive novelty is not certified. Reproduction compares analytic qubit responses with matrix exponentials, checks window endpoints and counts, evaluates piecewise constant higher-dimensional controls against their integrated diameter action, and checks exact binomial coverage in finite cases. Those checks are witnesses and regression safeguards, not premises of the general proofs. The offline demonstration varies the diameter cap, duration, response and unresolved generator axes; displayed candidates and additional-probe predictions are not experimental measurements. Failure criteria include an uncalibrated energy resource, clock or response; silent use of a mean energy in place of spectral diameter; endpoint division by zero; omitted revival windows; identifying a final state with its generator; unqualified confidence under adaptive or dependent samples; and an unjustified lift from generator compatibility to a complete world. Each would fail the stated interface. Next packages are: independently calibrate a finite apparatus and its control alphabet; treat open dynamics and clock uncertainty as new carriers; design finite probes that remove generator aliasing with controlled confidence. WR-VIII needs actual cosmological response maps and complete-realization assumptions; WR-IX needs accessible protocol and resource classes. This calculation opens neither branch. WREH remains its own programme and community, distinct from Boundary Compensation. Status: REVIEWABLE_DRAFT; no deposit, WR-VII DOI or journal-readiness assertion.

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